

Collision Theory

total no. of collision in rxn contains per unit time per unit vol. called collision freq. (Z)

* greater the value of activation energy slow will be ROR.

* Activation Energy is independent of temp.

* Activation Energy mainly depends upon -

→ Nature of reactants

→ Catalyst →

• +ve Catalyst for ROR by low Ea by providing a alternative path of low Energy to rxn

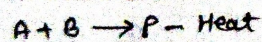
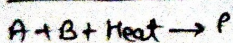
• Catalyst not change the value of any state funcⁿ that is $\Delta S, \Delta H, \Delta G, \Delta E$

• Catalyst can not start the rxn.

• Catalyst do not change K_{eqm} or E_{eq} path it change the time taken to reach E_{eq} .

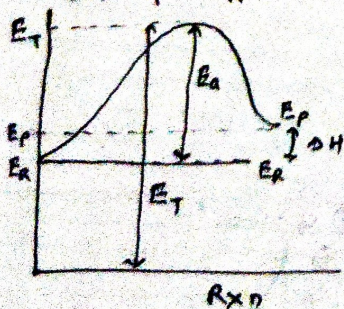
Exothermic

Exothermic

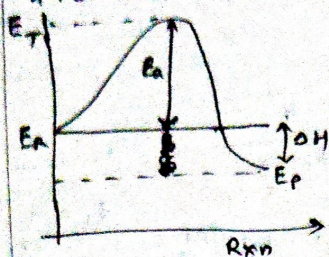
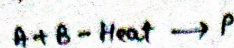
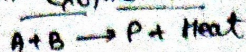


$$\Delta H, \Delta E = +ve$$

$$\Delta H = H_P - H_R = (+)$$

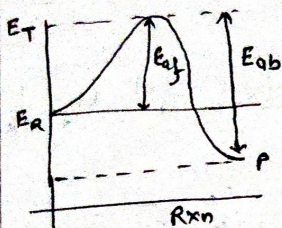
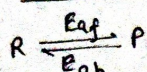


Exothermic



$$\Delta H = -ve$$

Reversible Rxn



$$E_{af} - E_{ab} = \Delta H$$

* for Endothermic Rxn the min. value of Ea

$$E_a > \Delta H$$

* for Exothermic Rxn the min. value of Ea

$$E_a \geq 0$$

Effect of catalyst or ROR

• Catalyst provide alternative path of lower threshold Energy & activation Energy. So, Rate of Rxn is ↑.

factor affecting Rate const (Arrhenius Eqⁿ)

$$K = A e^{-E_a/RT}$$

K depends on → Temp
- activation Energy
- Catalyst

E_a = Activation Energy

K = rate const

A = frequency factor / Arrhenius const

/ Pre-exponential factor

T = Temp (Kelvin)

E_a = J/mole, Cal/mole

R = Universal gas const

$$R = 8.3 \text{ J/mole K}$$

$$= 2 \text{ Cal/mole K}$$

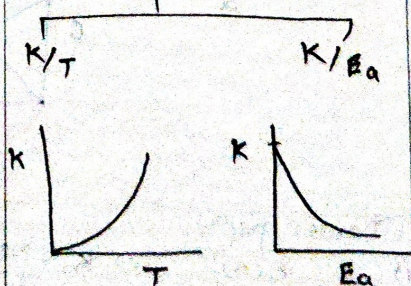
$e^{-E_a/RT}$ = Boltzmann const = fraction of molecules Activated

$$e^{-E_a/RT} = \text{unitless}$$

< 1

Unit of A = unit of K = time⁻¹

Graph



* Unit of freq. factor

$$\text{sec}^{-1}$$

* value of K

if $T \rightarrow \infty$

$$K = A$$

if $E = 0$

$$K = A$$

logarithm form of Rate const

$$① K = A e^{-E_a/RT}$$

$$\log_e K = \frac{-E_a}{RT} + \log_e A$$

$$\frac{d \ln K}{dT} = \frac{+E_a}{RT^2}$$

$$② P = A e^{-\Delta H_{vap}/RT}$$

$$\frac{d \ln P}{dT} = \frac{+\Delta H_{vap}}{RT^2}$$

$$③ K_c = A e^{-\Delta H_T/RT}$$

$$\frac{d \ln K_c}{dT} = \frac{+\Delta H_T}{RT^2}$$

$$④ K_w = A e^{-\Delta H_{ion}/RT}$$

$$\frac{d \ln K_w}{dT} = \frac{+\Delta H_{ion}}{RT^2}$$